

Capture the Flag

Your Student Portal is now dynamic thanks to interpolation.

But there's a problem.

Your announcement component currently has something like:

```
announcementTitle1 = "Enrollment is Now Open";  
announcementBody1 = "Enrollment for the next semester is now  
available.";
```

```
announcementTitle2 = "Intramurals 2026";  
announcementBody2 = "Registration is now open.";
```

```
announcementTitle3 = "Library Schedule";  
announcementBody3 = "The library will be open until 8 PM.";
```

And your HTML has to repeat the same card structure three times.

The school now has **20 announcements**.

Your mission:

Stop repeating yourself. Make Angular generate the cards for you.

Flag 0 – Mission Branch

Create and switch to a new branch: `feature/ngFor`



Capture

- Show your current Git branch.

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git branch
* feature/interpolation
  main
  my-new-branch
● PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git branch feature/ngFor
>>
⦿ PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git switch -c feature/ngFor
>>
fatal: a branch named 'feature/ngFor' already exists
● PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git checkout feature/ngFor
>>
Switched to branch 'feature/ngFor'
⦿ PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> |

Ln 42, Col 1 Spaces: 2
```

FLAG 1 — The Repetition Problem

Open your announcement component.

Look at your existing announcement cards.

You should currently have something similar to:

```
<div class="announcement-card">
  <h3>{{ announcementTitle1 }}</h3>
  <p>{{ announcementBody1 }}</p>
</div>
```

```
<div class="announcement-card">
  <h3>{{ announcementTitle2 }}</h3>
  <p>{{ announcementBody2 }}</p>
</div>
```

```
<div class="announcement-card">
  <h3>{{ announcementTitle3 }}</h3>
  <p>{{ announcementBody3 }}</p>
</div>
```

Mission

Imagine the school gives you **10 more announcements**.

Would you want to copy and paste the card **10 more times**?

Don't change anything yet.



Capture: Your repeated announcement cards.

```
profile-card.css  announcement-card.html  TS announcement-card.ts  MyFirstAngularProject

src > app > components > announcement-card > announcement-card.html > HTML Language Features > div.ann

Go to component
1  <div class="announcement-card">
2    <h2 class="announcement-title">{{ announcementTitle1 }}</h2>
3    <p class="Announcement-content">{{ announcementBody1 }}</p>
4  </div>
5
6  <div class="announcement-card">
7    <h2 class="announcement-title">{{ announcementTitle2 }}</h2>
8    <p class="Announcement-content">{{ announcementBody2 }}</p>
9  </div>
10
11 <div class="announcement-card">
12   <h2 class="announcement-title">{{ announcementTitle3 }}</h2>
13   <p class="Announcement-content">{{ announcementBody3 }}</p>
14 </div>
15
16 <div class="announcement-card">
17   <h2 class="announcement-title">{{ announcementTitle4 }}</h2>
18   <p class="Announcement-content">{{ announcementBody4 }}</p>
19 </div>
20
21 <div class="announcement-card">
22   <h2 class="announcement-title">{{ announcementTitle5 }}</h2>
23   <p class="Announcement-content">{{ announcementBody5 }}</p>
24 </div>
25
26 <div class="announcement-card">
27   <h2 class="announcement-title">{{ announcementTitle6 }}</h2>
28   <p class="Announcement-content">{{ announcementBody6 }}</p>
29 </div>
30
31 </div>
```

```
src > app > components > announcement-card > announcement-card.html > HTML Language F

24 </div>
25
26 <div class="announcement-card">
27   <h2 class="announcement-title">{{ announcementTitle6 }}</h2>
28   <p class="Announcement-content">{{ announcementBody6 }}</p>
29 </div>
30
31 <div class="announcement-card">
32   <h2 class="announcement-title">{{ announcementTitle7 }}</h2>
33   <p class="Announcement-content">{{ announcementBody7 }}</p>
34 </div>
35
36 <div class="announcement-card">
37   <h2 class="announcement-title">{{ announcementTitle8 }}</h2>
38   <p class="Announcement-content">{{ announcementBody8 }}</p>
39 </div>
40
41 <div class="announcement-card">
42   <h2 class="announcement-title">{{ announcementTitle9 }}</h2>
43   <p class="Announcement-content">{{ announcementBody9 }}</p>
44 </div>
45
46 <div class="announcement-card">
47   <h2 class="announcement-title">{{ announcementTitle10 }}</h2>
48   <p class="Announcement-content">{{ announcementBody10 }}</p>
49 </div>
```

FLAG 2 — Put the Data Together

Instead of creating separate variables, create **one array**.

Research Hint

An array can contain multiple objects.

You don't need to memorize the syntax.

Search:

Angular array of objects

 Capture: Your **announcements** array.

```
src > app > components > announcement-card > ts announcement-cards > Announcementcard > announcements > body
1  import { Component } from '@angular/core';
2
3  @Component({
4    selector: 'app-announcement-card',
5    standalone: true,
6    templateUrl: './announcement-card.html',
7    styleUrls: ['./announcement-card.css']
8  })
9  export class Announcementcard {
10
11    announcements = [
12      {
13        title: 'Enrollment Reminder',
14        body: 'Enrollment for the first Semester AY 2026-2027 will end on August 08, 2026. Please complete your requirements on time.'
15      },
16      {
17        title: 'Official ID Picture Taking',
18        body: 'Official ID Picture Taking for all enrolled students will be held on August 20, 2026 at the Don Bosco AVR from 8:00 AM to 4:00 PM. Students are r
19      },
20      {
21        title: 'Intramurals 2026',
22        body: 'Registration for the annual Intramurals is now open. Students may register for sports and cultural events until September 05, 2026.'
23      },
24      {
25        title: 'SDP - CLUB MEMBER SCREENING',
26        body: 'Instructions for the Club Member Screening will be given after the holy mass.'
27      },
28      {
29        title: 'CLASS SUSPENSION',
30        body: 'Due to inclement weather, all face-to-face classes are suspended today.'
31      },
32      {
33        title: 'BUWAN NG WIKA: BARRIO FIESTA CELEBRATION',
34        body: 'Join us as we celebrate Buwan ng Wika with a fun filled Barrio Fiesta!'
35      },
36    ]
37  }
```

```
    {
      title: 'CLASS SUSPENSION',
      body: 'Due to inclement weather, all face-to-face classes are suspended today.'
    },
    {
      title: 'BUWAN NG WIKA: BARRIO FIESTA CELEBRATION',
      body: 'Join us as we celebrate Buwan ng Wika with a fun filled Barrio Fiesta!'
    },
    {
      title: 'EXAMINATION NOTICE',
      body: 'Midterm examination will be on September 14, 2026.'
    },
    {
      title: 'LIBRARY ADVISORY',
      body: 'Students are reminded to return borrowed books on or before the due date to avoid overdue fines.'
    },
    {
      title: 'LOST AND FOUND',
      body: 'A black wallet was found near the school cafeteria. Owner may claim it at the SAO office by presenting proper Identification and descri
    },
    {
      title: 'BOSCONIAN NIGHT NOTICE',
      body: 'Get ready for an exciting and unforgettable evening as we celebrate Bosconian Night 2026!'
    }
  ]
}
```

FLAG 3 — Meet `*ngFor`

Now comes the important part.

You have:

```
announcements = [  
  { ... },  
  { ... },  
  { ... }  
];
```

How can Angular create one card **for every item** in that array?

Hint

Research:

**Angular ngFor*

You should discover syntax similar to:

```
*ngFor="let announcement of announcements"
```

Your mission is to use it on your announcement card.

Instead of having:

```
<div class="announcement-card">
```

repeated three times, you should have **one card template** that Angular repeats.



Capture:

- Your updated code.

```
src > app > components > announcement-card > announcement-card.html > HTML Language Features >
Go to component
1 <div class="announcement-card" *ngFor="let announcement of announcements">
2
3   <h2 class="announcement-title">
4     {{ announcement.title }}
5   </h2>
6
7   <p class="Announcement-content">
8     {{ announcement.body }}
9   </p>
10
11 </div>
```

```
src > app > components > announcement-card > TS announcement-cards.ts > Announcementcard > announcements
1 import { Component } from '@angular/core';
2 import { CommonModule } from '@angular/common';
3
4 @Component({
5   selector: 'app-announcement-card',
6   standalone: true,
7   imports: [CommonModule],
8   templateUrl: './announcement-card.html',
9   styleUrls: ['./announcement-card.css']
10 })
11 export class Announcementcard {
12
13   announcements = [
14     {
15       title: 'Enrollment Reminder',
16       body: 'Enrollment for the First Semester AY 2026-2027 will end on August 08, 2026. Please complete your requirements on time.'
17     },
18     {
19       title: 'Official ID Picture Taking',
20       body: 'Official ID Picture Taking for all enrolled students will be held on August 20, 2026 at the Don Bosco AVR from 8:00 AM to 4:00 PM. Students must bring a recent 2x2 inch photo.'
21     },
22     {
23       title: 'Intramurals 2026',
24       body: 'Registration for the annual Intramurals is now open. Students may register for sports and cultural events until September 05, 2026.'
25     },
26     {
27       title: 'SDP - CLUB MEMBER SCREENING',
28       body: 'Instructions for the Club Member Screening will be given after the holy mass.'
29     },
30     {
31       title: 'CLASS SUSPENSION',
32       body: 'Due to inclement weather, all face-to-face classes are suspended today.'
33     },
34     {
35       title: 'BUWAN NG WIKAI: BARRIO FIESTA CELEBRATION',
36       body: 'Join us as we celebrate Buwan ng Wika with a fun filled Barrio Fiesta!
```

```
src > app > components > announcement-card > TS announcement-cards.ts > Announcementcard > announcements
11 export class Announcementcard {
12
13   announcements = [
14     {
15       title: 'Intramurals 2026',
16       body: 'Registration for the annual Intramurals is now open. Students may register for sports and cultural events until September 05, 2026.'
17     },
18     {
19       title: 'SDP - CLUB MEMBER SCREENING',
20       body: 'Instructions for the Club Member Screening will be given after the holy mass.'
21     },
22     {
23       title: 'CLASS SUSPENSION',
24       body: 'Due to inclement weather, all face-to-face classes are suspended today.'
25     },
26     {
27       title: 'BUWAN NG WIKAI: BARRIO FIESTA CELEBRATION',
28       body: 'Join us as we celebrate Buwan ng Wika with a fun filled Barrio Fiesta!'
29     },
30     {
31       title: 'EXAMINATION NOTICE',
32       body: 'Midterm examination will be on September 14, 2026.'
33     },
34     {
35       title: 'LIBRARY ADVISORY',
36       body: 'Students are reminded to return borrowed books on or before the due date to avoid overdue fines.'
37     },
38     {
39       title: 'LOST AND FOUND',
40       body: 'A black wallet was found near the school cafeteria. Owner may claim it at the SAO office by presenting proper identification and description.'
41     },
42     {
43       title: 'BOSCONIAN NIGHT NOTICE',
44       body: 'Get ready for an exciting and unforgettable evening as we celebrate Bosconian Night 2026!'
45     }
46   ];
47 }
```

FLAG 4 — Display the Current Announcement

You now have a loop variable.

Inside the repeated card, display title and body:

Your browser should now display all eight announcements.

Important

You should have:

1 array

1 card template

8 displayed cards

Not:

8 arrays

8 card templates



Capture:

- Browser showing all announcements.
- Updated code.

Don Bosco Student Portal



Jownie Ann D. Malate

Student ID: 23017616

Courses: Bachelor of Science in Information Technology

Year Level: 4th Year

Section: N/A

Nickname: YANG-YANG

Email: jowniemalate@dbtc-cebu.edu.ph

Announcements!

Stay updated with the latest school information.

Enrollment Reminder

Enrollment for the First Semester AY 2026-2027 will end on August 06, 2026. Please complete your requirements on time.

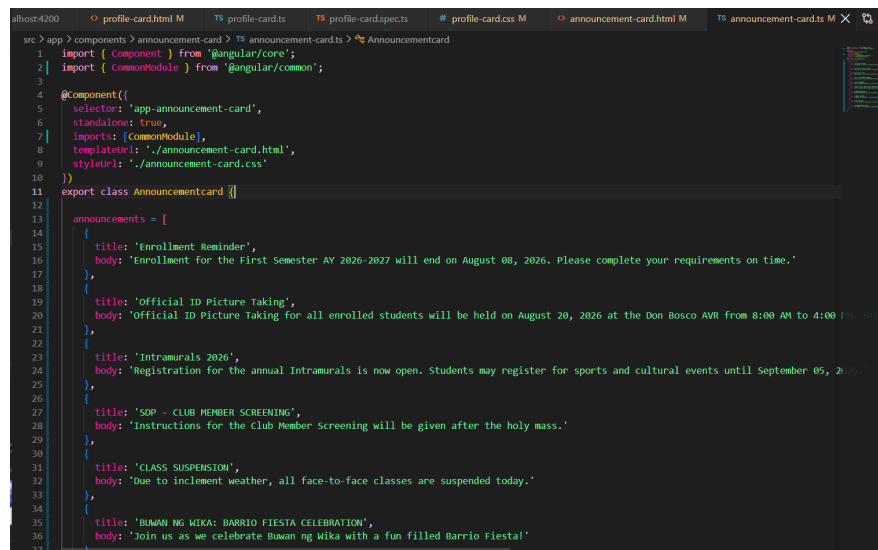
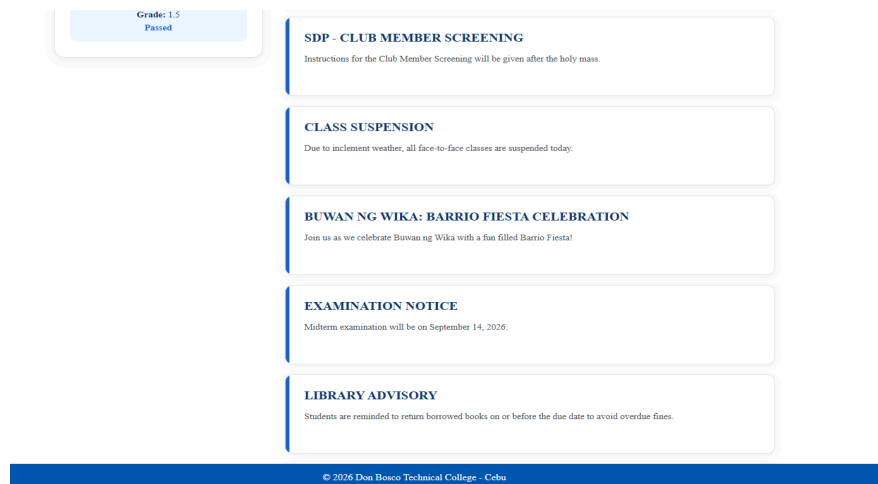
Official ID Picture Taking

Official ID Picture Taking for all enrolled students will be held on August 20, 2026 at the Don Bosco AVR from 8:00 AM to 4:00 PM. Students are required to wear their complete school uniform and present their registration form.

Intramurals 2026

Registration for the annual Intramurals is now open. Students may register for sports and cultural events until September 05, 2026.

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FLAG 5 — Add More Data

Now test whether your loop actually works.

Add **two more announcements** to the array.

You should **not change your HTML**.

If two additional cards automatically appear:

 **FLAG CAPTURED**

 **Capture:** Browser showing all ten announcements.



Jownie Ann D. Malate

Student ID: 23017636

Course: Bachelor of Science in Information Technology

Year Level: 4th Year

Section: N/A

Nickname: YANIE-YANG

Email: jownieann.malate@dbu-ortho.edu.ph

Grade: 1.5
Passed

Announcements!

Stay updated with the latest school information.

Enrollment Reminder

Enrollment for the First Semester AY 2026-2027 will end on August 06, 2026. Please complete your requirements on time.

Official ID Picture Taking

Official ID Picture Taking for all enrolled students will be held on August 28, 2026 at the Don Bosco AVR from 8:00 AM to 4:00 PM. Students are required to wear their complete school uniform and present their registration form.

Intramurals 2026

Registration for the annual Intramurals is now open. Students may register for sports and cultural events until September 05, 2026.

SDP - CLUB MEMBER SCREENING

Instructions for the Club Member Screening will be given after the holy mass.

CLASS SUSPENSION

Due to inclement weather, all face-to-face classes are suspended today.

BUWAN NG WIKI: BARRIO FIESTA CELEBRATION

Join us as we celebrate Buwan ng Wiki with a fun filled Barrio Fiesta!

EXAMINATION NOTICE

Midterm examination will be on September 14, 2026.

LIBRARY ADVISORY

Students are reminded to return borrowed books on or before the due date to avoid overdue fines.

LOST AND FOUND

A black wallet was found near the school cafeteria. Owner may claim it at the SAC office by presenting proper identification and describing its contents.

BOSCONIAN NIGHT NOTICE

Get ready for an exciting and unforgettable evening as we celebrate Bosconian Night 2026!

FLAG 6 — Make the Cards More Realistic

Your announcement objects currently contain only:

title
body

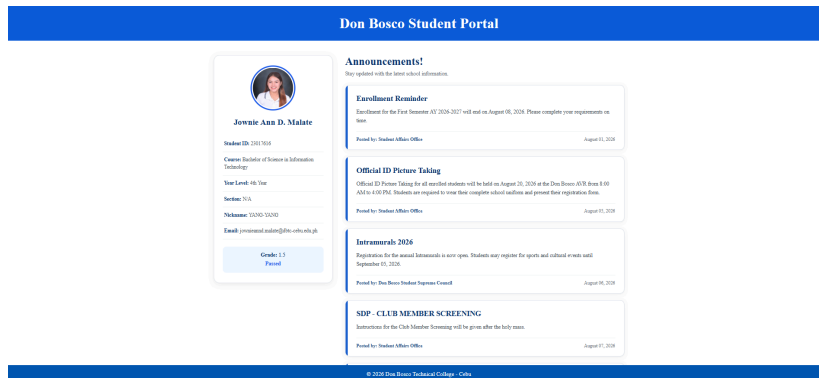
Add more information: author and date.

Update your card to display all fields.

Use **interpolation** for all values.



Capture: Your richer announcement cards.



FLAG 7 — Student List

Announcements aren't the only thing that can be repeated.

Create another array named students.

Create a simple student card.

Use ***ngFor** to display every student.

Expected example:

Juan Dela Cruz
BSIT • 2nd Year

Maria Santos
BSIT • 3rd Year

Pedro Garcia
BSCS • 1st Year

 **Capture:** Student list.

Students

Student directory

Mary Beth D. Gracia
BSIT • 4th Year

Bonna Mae L. Pitogo
BSIT • 4th Year

Niel Jhon E. Celocia
BSIT • 4th Year

FLAG 8 — One Template, Many Students

Add another student to your array.

Do **not** modify the HTML.

The new student should automatically appear.

This is the important concept:

DATA CHANGES



ARRAY CHANGES



ANGULAR GENERATES THE UI

 **Capture:** Before and after adding the student.

Before:

Students

Student directory

Mary Beth D. Gracia

BSIT • 4th Year

Bonna Mae L. Pitogo

BSIT • 4th Year

Niel Jhon E. Celocia

BSIT • 4th Year

After:

Students

Student directory

Mary Beth D. Gracia

BSIT • 4th Year

Bonna Mae L. Pitogo

BSIT • 4th Year

Niel Jhon E. Celocia

BSIT • 4th Year

Kurt Wojtyle S. Rizal

BSIT • 4th Year

FLAG 9 — Research Challenge

I will **not** provide the next syntax.

Research:

**Angular ngFor index*

Find out how to access the position of the current item.

Use it to display:

#1 Juan Dela Cruz

#2 Maria Santos

#3 Pedro Garcia

Without hardcoding the numbers.

Hint:

let i = index



Capture:

- Your numbered student list.
- Your code

Students

Student directory

#1 Mary Beth D. Gracia

BSIT • 4th Year

#2 Bonna Mae L. Pitogo

BSIT • 4th Year

#3 Niel Jhon E. Celocia

BSIT • 4th Year

#4 Kurt Wojtyle S. Rizal

BSIT • 4th Year

```
src > app > components > student-list > student-list.html > HTML Language f
Go to component
1  <div class="student-list">
2
3  <div
4    class="student-card"
5    *ngFor="let student of students; let i = index"
6  >
7
8    <h2 class="student-name">
9      | #{{ i + 1 }} {{ student.name }}
10    </h2>
11
12    <p class="student-details">
13      | {{ student.course }} • {{ student.yearLevel }}
14    </p>
15
16  </div>
17
18 </div>
```

FLAG 10 — Empty List

Now test what happens when there are no announcements.

Change:

```
announcements = [];
```

What happens?

Write a short observation in your documentation:

What happens when the array is empty?

Answer: When I removed the announcement array, no announcement card appear on my browser but the other parts of the browser like the profile card and the students list still remains.

Then restore your announcements.

 **Capture:** Your experiment and observation.

Don Bosco Student Portal



Jownie Ann D. Malate

Student ID: 23017616

Course: Bachelor of Science in Information Technology

Year Level: 4th Year

Section: NA

Nickname: YANG-YANG

Email: jownieann.d.malate@dbtc-cebu.edu.ph

Grade: 1.5
Passed

Announcements

Stay updated with the latest school information.

Students

Student directory

#1 Mary Beth D. Gracia
BSIT - 4th Year

#2 Donna Mae L. Pitogo
BSIT - 4th Year

#3 Niel Jhon E. Celocia
BSIT - 4th Year

#4 Kurt Wojtyla S. Rizal
BSIT - 4th Year

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FLAG 11 — Dynamic Student Portal

Your Student Portal should now contain:

 **Your Student Profile**

Still using interpolation:

Name
Student ID
Course
Year
Grade
Academic Standing
Email

Announcements

Generated from:

announcements = [...]

Using:

*ngFor

Student List

Generated from:

students = [...]

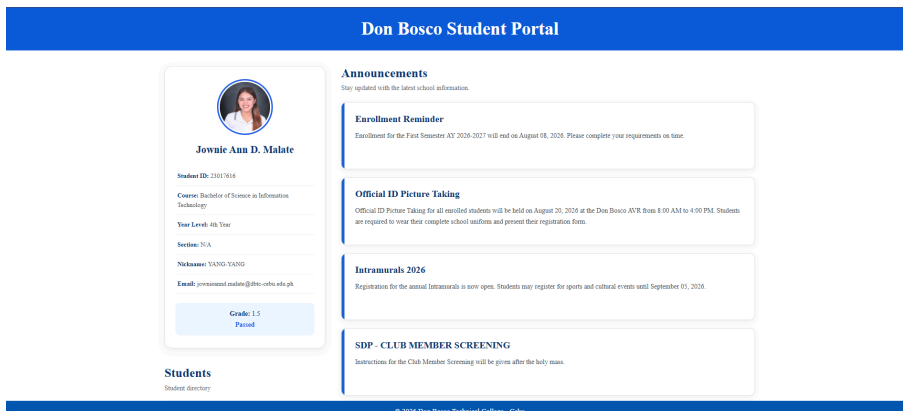
Using:

*ngFor and index

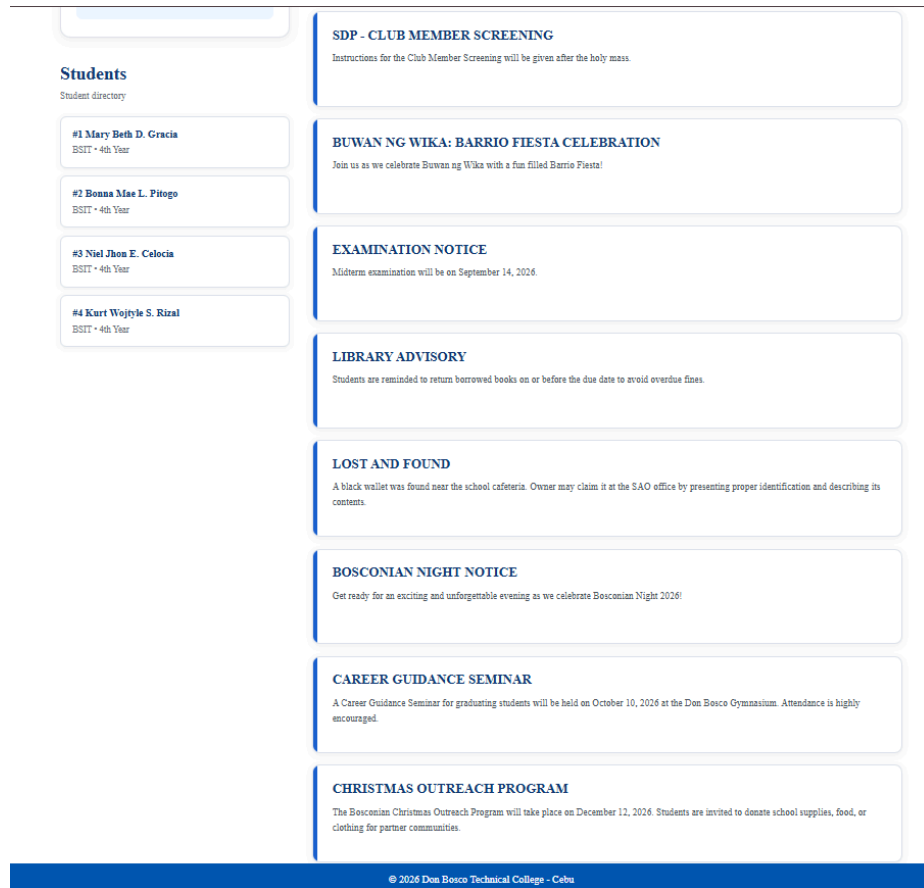
Your final requirement

Final Screenshot

- Entire browser UI of all requirements in this activity



The screenshot displays the Don Bosco Student Portal. On the left, a student profile for Jowale Ann D. Malafe is shown, including her photo, student ID (23057616), course (Bachelor of Science in Information Technology), year level (4th Year), section (N/A), nickname (JANNI), and email (jannimalefe@dbcs-ccbo.edu.ph). Below the profile is a 'Students' link. The main content area is titled 'Announcements' and lists four items: 'Enrollment Reminder' (due August 08, 2026), 'Official ID Picture Taking' (August 20, 2026), 'Intramurals 2026' (registration until September 03, 2026), and 'SDP - CLUB MEMBER SCREENING' (instructions to be given after the holy mass). The footer indicates the portal is for Don Bosco Technical College - Cebu, © 2020.



FLAG 12 — Commit & Push

Final Screenshot

- Successful push to feature/ngFor

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git status
no changes added to commit (use "git add" and/or "git commit -a")
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git add .
warning: in the working copy of 'src/app/components/announcement-card/announcement-card.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/profile-card/profile-card.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/student-list/student-list.html', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/student-list/student-list.spec.ts', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/app/components/student-list/student-list.ts', LF will be replaced by CRLF the next time Git touches it
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git commit -m "Dynamic Student Portal"
[feature/ngfor 476b070] Dynamic Student Portal
12 files changed, 714 insertions(+), 182 deletions(-)
create mode 100644 src/app/components/student-list/student-list.css
create mode 100644 src/app/components/student-list/student-list.html
create mode 100644 src/app/components/student-list/student-list.spec.ts
create mode 100644 src/app/components/student-list/student-list.ts
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git push origin main
Everything up-to-date
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git status
On branch feature/ngfor
nothing to commit, working tree clean
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git branch
  feature/components
  feature/interpolation
* feature/ngfor
  main
  my-new-branch
PS C:\Users\JOWNIE\OneDrive\Documents\my-first-angular-project> git push -u origin feature/ngfor
Enumerating objects: 34, done.
Counting objects: 100% (34/34), done.
Delta compression using up to 8 threads
Compressing objects: 100% (20/20), done.
Writing objects: 100% (20/20), 5.95 KiB | 243.00 KiB/s, done.
Total 20 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 4 local objects.
remote:
remote: Create a pull request for 'feature/ngfor' on GitHub by visiting:
  https://github.com/JOWNIE/my-first-angular-project/pull/new/feature/ngfor
```